C6 Internal Resistance

C6.1 A battery with an e.m.f. of 12.0 V is supplying 50 A at 10.5 V to a motor. Calculate the internal resistance of the battery.

C6.2 A power station generator with an e.m.f. of 22 kV has an internal resistance of 0.90 mΩ. The current is 22 kA. Calculate the difference between the e.m.f. and the terminal p.d.

C6.3 According to its data sheet, a portable solar cell for camping has an 'open circuit voltage' of 25.6 V. This means that the terminal p.d. is 25.6 V when no load is connected. It also has a 'short circuit current' of 2.06 A. This means that if the terminals are connected together directly a current of 2.06 A flows.

a) State the e.m.f. of the solar cell.

b) Calculate the internal resistance of the solar cell.

Warning: never measure the short circuit current of a supply component unless you have been assured by a qualified professional that it is safe to do so, and that you have the correct equipment.

Answers

C6.1 $\frac{12-10.5}{50}$ =0.030 Ω

C6.2 $22 \times 10^3 \times 0.9 \times 10^{-3}$ =19.8 V

C6.3

a) 25.6 V

b) $\frac{25.6}{2.06}$ =12.427 Ω